Active Maintenance Report: murphy_etal_2026

Murphy, Coppock, Hill, and Vavreck (2026, Political Communication): Do Emotional Ads Persuade? Evidence from Real-Time Campaign Advertising Experiments

Table of contents

1 Summary 2

2 Paper overview 3

3 Original archive reproducibility 3

4 Errata 4

4.1 Read and not filed 4

5 Ground truth 5

6 The extraction and the instruments 5

7 Maintained rewrite 6

8 Figure verification 6

9 R environment 6

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1 Summary

This repository reruns the deposit of an article on whether the emotional content of campaign ads predicts their persuasive effects, compares the article and its appendix with the rerun, and lists the errors found.

- **The deposit runs.** All 27 of its scripts finish without error, both in place and after every derived file is removed.
- **The page matches the deposit.** 41 of 42 extracted numbers agree at the printed precision, and so do all 966 appendix table cells checked. The one disagreement is the two-ad sample size.
- **Matching the deposit does not make the numbers right.** Sanders favorability is rescaled twice in the deposit's cleaning script, which the article prints faithfully. See Errata.
- **The article's central null holds under every correction computed here.** None of the 48 refits of the Figure 2, 3, and 4 slopes reaches $p < 0.05$. These are the corrections listed in the errata and not all conceivable ones.
- **This repository holds no independent rewrite.** Every `value_rewrite` comes from rerunning the deposit's own code, so an error in that code that is not reported below would pass here too.

Link	
Article	10.1080/10584609.2026.2726266
Replication archive	10.7910/DVN/CB3LP1
Errata	`murphy_etal_2026_errata.pdf` in this repository

Layout. `original/` holds the deposit's zip once fetched and `original_extracted/` its contents; neither is redistributed. `maintained/` holds the rerun, the comparison, and the refits. `ground_truth/` holds the extraction from the published page and the comparison with the rerun. `errata.qmd` is the source of the errata note, and `errata_entries.csv` is its machine-readable copy.

To reproduce. Download the repository, open `murphy_etal_2026.Rproj`, and run `source("run_all.R")`. The stripped rerun and the accuracy rerun each took between 10 and 15 minutes here. Set the environment variable `MURPHY_REUSE_ARCHIVE_RUN=TRUE` to reuse a finished one.

2 Paper overview

The article reports 29 waves of survey experiments fielded by YouGov during the 2016 presidential campaign, in which subjects saw a campaign ad or a placebo. It regresses each ad's conditional average treatment effect on favorability, vote choice, turnout, and issue support on subjects' average emotional reactions to that ad, and finds slopes that are close to zero.

3 Original archive reproducibility

The deposit is 1 zip with 107 members, all of which match the manifest on MD5 and byte size. It ran in two passes.

Pass	Starting point	Scripts OK	Scripts failed
In place	The deposit as shipped	27	0
Stripped	Every file in `output/`, `data/estimates/`, and both cleaned data files removed	27	0

Each script's outcome in both passes is in `ground_truth/archive_run_status.csv`, and what the stripped pass did to each file of the deposit is in `ground_truth/archive_overwrites.csv`. The in-place pass overwrote every derived file, which leaves six input files as the only ones older than the run. That pass was run once from an unzipped copy, and its log is `maintained/output/archive_inplace_run.log`. `run_all.R` repeats only the stripped pass.

The stripped rerun's 60 output files compare with the deposited ones as follows.

verdict	files
byte-identical	28
differs	3
environment record, not compared	1
identical but for its comment header	1
identical but for timestamp and R version	13
numbers differ by at most measure	4
share of channel values that differ is measure	10

- The four CSVs that differ do so by at most 1.6e-09, which is numerical noise.

- Three PDFs differ beyond their timestamp and R version: Figure 1, Figure A.2, and Figure E.7. Figures A.2 and E.7, and their PNGs, use `position_jitter` with no seed, so they differ from one run to the next. Figure 1 places its ad labels with `ggrepel` and no seed, so its PDF can differ from one run to the next; in the run of 2026-11-12 it differed in one text coordinate by 0.01 points. The other PNGs differ in at most 0.01 percent of channel values, from rasterization, and their PDFs agree.

The deposit's audit script ends with this line:

> CLAIM SUMMARY: asserted=72 matched=67 mismatched=0 unasserted=6

Read it with care. "Asserted" counts every row, including the 5 that its source marks `expect = "derived"`, which are printed and never compared. Those rows and one evidence line make up "unasserted", which is why asserted exceeds matched with nothing mismatched. The two-ad sample size is one of those rows, so the summary cannot fail on it.

4 Errata

The note lists 13 entries: 3 corrected quantities, 7 brief corrections to the text, and 3 notes that correct nothing because neither the page nor the deposit settles what was intended.

Entry	Item	Form
1	Sanders favorability is rescaled twice	full
2	The two-advertisement sample	full
3	Turnout and issue estimates take the sign of the ad's valence	full
4	Note 5 on the primary ads	note
5	Three attack ads are coded positive	note
6	Whose favorability	brief
7	The border item	brief
8	The note under Figure E.10	brief
9	Two ads named Sacrifice	note
10	The estimator is weighted	brief
11	A dropped word	brief
12	A missing construction	brief
13	Spelling and number	brief

4.1 Read and not filed

These were read and left out of the errata. They are recorded so that a later reader need not rediscover them.

- **"Each week" (p. 4).** The deposit has 29 dated waves spanning 34 weeks, with six 14-day gaps. The sentence describes what happened in a week with a wave, and "29 weeks of experiments" counts those waves correctly, so no text is wrong.
- **`will_vote.` `clean_exps.R` codes `match_DTHC` 1 and 2 as 1 and 9 as 0, and leaves code 8 missing, which is 748 raw responses.** The deposit holds no codebook to name that code, and the article prints no statement about it, so there is nothing on the page to correct.
- **"CORRECTED AT PROOF" comments in `in_text_claims.R`.** They sit beside stale quotations of earlier wording. The values checked here agree with the published PDF, so this concerns the deposit's comments and not the article.

5 Ground truth

`ground_truth/murphy_etal_2026_ground_truth.csv` holds 45 claims: 42 numbers compared with the rerun and 3 descriptive statements given a computed verdict. `value_paper` is what the page prints, and `value_rewrite` is the rerun's value. `value_script` repeats it, because the rerun is the deposit's own code. Rows the errata reports carry a `defect_locus` and a `disposition`. The descriptive verdicts are 0 holding and 3 not holding.

6 The extraction and the instruments

`published_claims.csv` covers the slopes printed in Figures 2, 3, and 4, the Figure E.10 slope, and the counts and reliabilities in the text. It leaves uncovered most of the matchup percentages and wave sizes that the deposit's own audit checks. `published_appendix_values.csv` and `published_table_d16.csv` cover the appendix's regression tables and its ad-coding table.

The instruments are the comparison in `build_ground_truth.R`, `maintained/in_text_claims.R`, which checks every extracted claim against the rerun, the corrected refits, and the errata spine with `excheckr` and stops `run_all.R` on any failing verdict, and the deposit's own `in_text_claims.R`, which is reported above. Neither can detect an error that the page and the code share, and the largest entries in the errata are of that kind. They came from reading the code and from a range table of the cleaned variables.

`build_ground_truth.R` halts on a broken join, a claim with no value or verdict, a spine id with no extraction row, a number that disagrees with the page and is absent from the spine, a `paper_internal` row with no disposition, and any disagreeing appendix cell.

Float	Published numbers	Agreeing
Regression tables	918	918
Table D.16	48	48

`ground_truth/negative_controls.R` breaks one input at a time and requires the matching gate to stop the build. All 16 of 16 runs behave as required. These runs show that the gates fire on these faults and say nothing about faults not tried.

7 Maintained rewrite

There is none. The deposit is recent code in the same style this project would write, so this repository reruns it and adds `maintained/corrected_slopes.R`, which refits the published meta-regressions on the deposit's own estimates under the corrections. That script's published specification returns every printed cell of Tables C.1, C.6, C.11, and E.17 before it applies a correction. `maintained/corrected_accuracy.R` reruns the deposit's two accuracy bootstraps with the duplicated Sanders line deleted and the deposit's seed, since those results are not linear in the CATEs. An independent rewrite remains undone, and it is what could catch a further error in the estimation code.

8 Figure verification

Figures are compared through the rerun's files in the table above, and through the slope labels on Figures 2, 3, and 4 in the ground truth. Nobody compared the published figures with the rerun's by eye.

9 R environment

	Deposit	This rerun
R	R version 4.6.0 (2026-04-24)	R version 4.6.2 (2026-10-29)
estimatr	NA	NA

The differences above are the only drift measured between these environments.